Master of Computer Applications

Semester – III

Data Analytics for Business

Experiment No.: 7

Title: Developing finance KPI dashboards

Submitted By:

Name: Lavanya E

Reg. No.: 2511022250083

Date: 15- Aug, 2026

Submitted To:

Faculty Name: Mr. Aman Kumar Sharma

Department of Computer Application

Alliance University

Chandapura - Anekal Main Road, Anekal

Bengaluru – 562106

Developing Finance KPI Dashboard

Problem Statement

A manufacturing company wants to monitor its overall financial performance using an interactive Power BI dashboard. Senior management requires a single dashboard to analyze budget allocation, actual expenditure, budget variance, budget utilization, department performance, category-wise spending, regional performance, and payment methods. The objective is to design an interactive Finance KPI Dashboard using the Budget vs Actual Financial Dataset that supports effective financial monitoring and business decision-making.

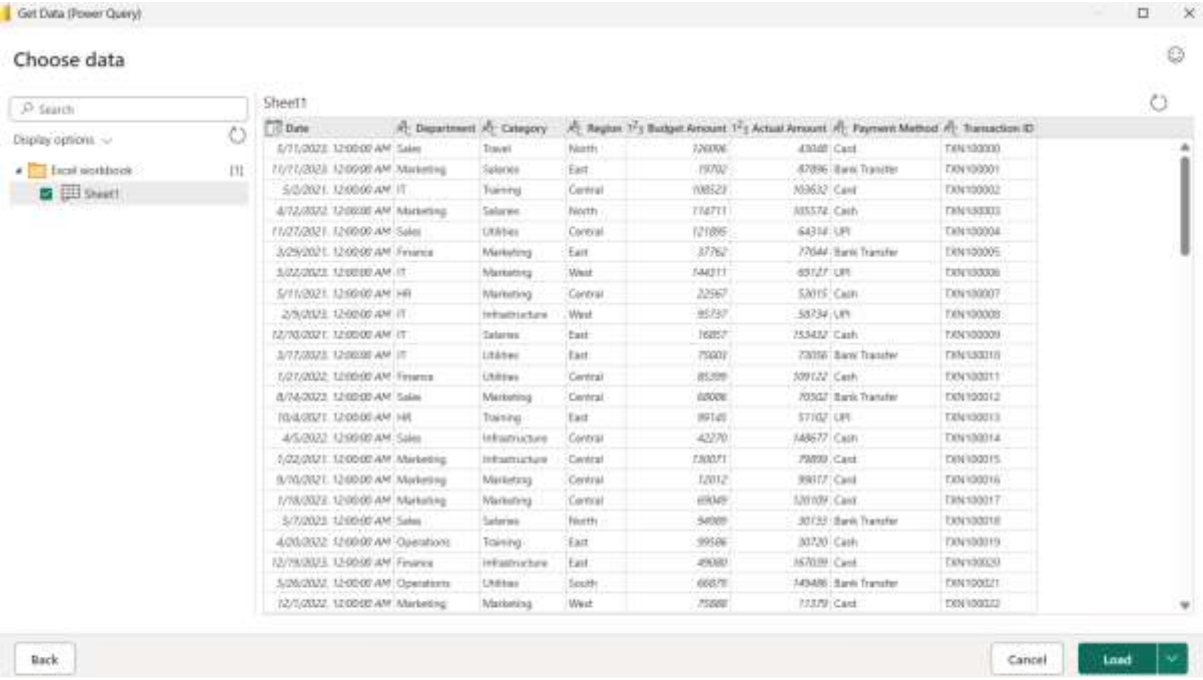
Dataset Used: Budget vs Actual Financial Dataset (Kaggle)

Methodology / Steps Followed

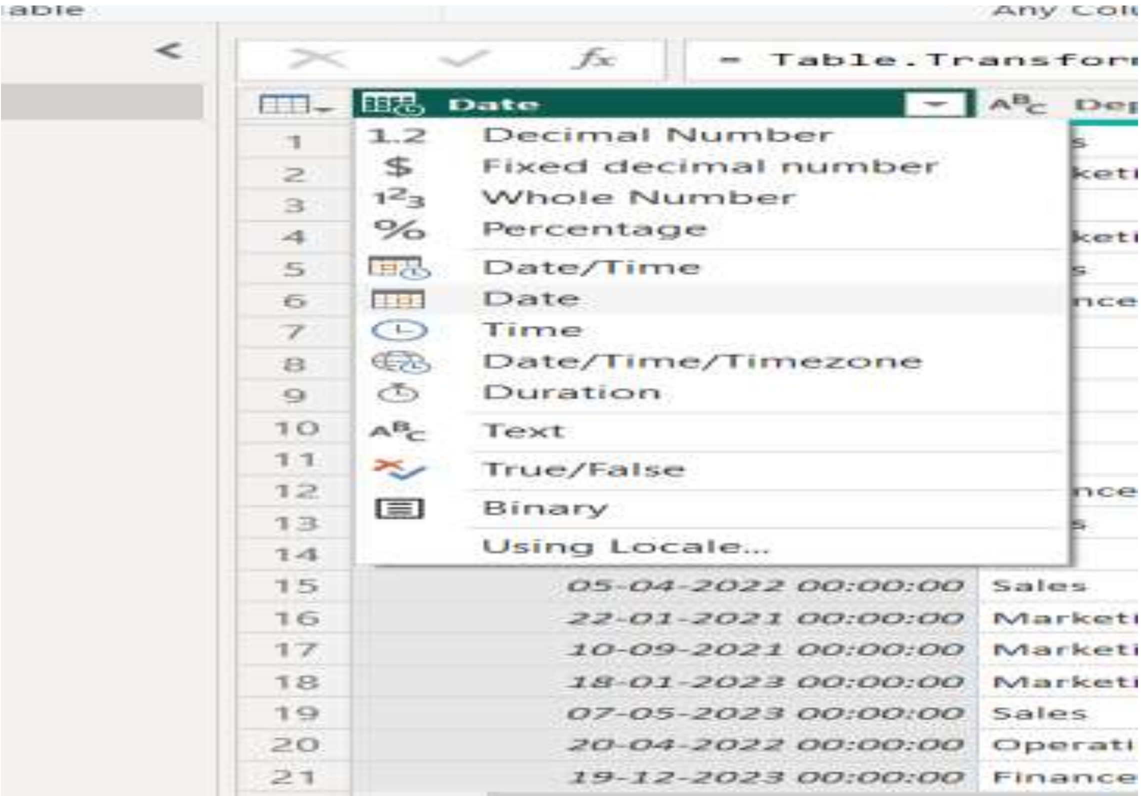
1. **Downloaded** the Budget vs Actual Financial Dataset from Kaggle.
2. **Imported** the dataset into Power BI Desktop using the **Get Data** option.
3. **Cleaned and verified** the data using Power Query by checking data types, duplicate records, and missing values.
4. **Converted and verified** the Date, Budget Amount, and Actual Amount columns into the appropriate data types.
5. **Created financial KPIs** to analyze Total Budget, Total Actual, Budget Variance, Budget Utilization, and Total Transactions.
6. **Created KPI Card Visuals** to display Total Budget, Total Actual, Budget Variance, Budget Utilization %, and Total Transactions.
7. **Created charts** to compare Budget and Actual Amount by Department, Category, Region, and Year.
8. **Created additional visualizations** such as Payment Method analysis, Budget Variance by Department, and Budget Amount by Category.
9. **Added interactive slicers** for Year, Department, Region, Category, and Payment Method to enable customized analysis.
10. **Applied professional formatting** using a dark corporate theme, contrasting colors, consistent fonts, and proper alignment to improve dashboard readability and presentation.

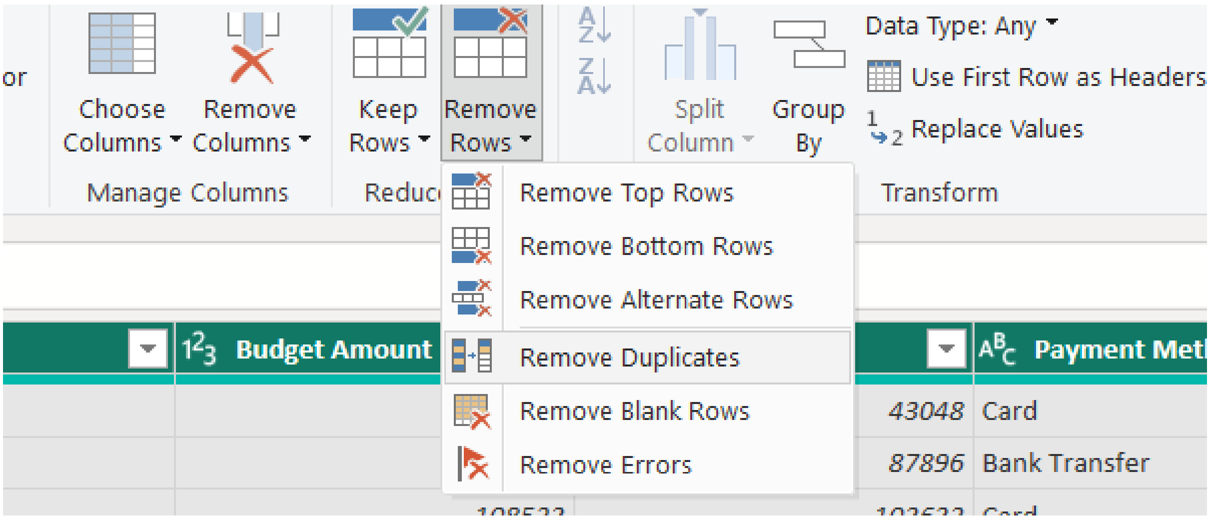
Output

➤ Loading the Budget vs Actual Financial Dataset



➤ Data Cleaning and Data Type Verification





➤ Creating Financial KPI Measures using DAX

```
Date Table =
ADDCOLUMNS(
    CALENDAR(
        MIN('Financial Data'[Date]),
        MAX('Financial Data'[Date])
    ),
    "Year", YEAR([Date]),
    "Month Number", MONTH([Date]),
    "Month", FORMAT([Date], "MMM"),
    "Year Month", FORMAT([Date], "YYYY-MM")
)
```

```
Total Budget =
SUM('Financial Data'[Budget Amount])
```

```
Total Actual =
SUM('Financial Data'[Actual Amount])
```

```
Budget Variance =
[Total Budget] - [Total Actual]
```

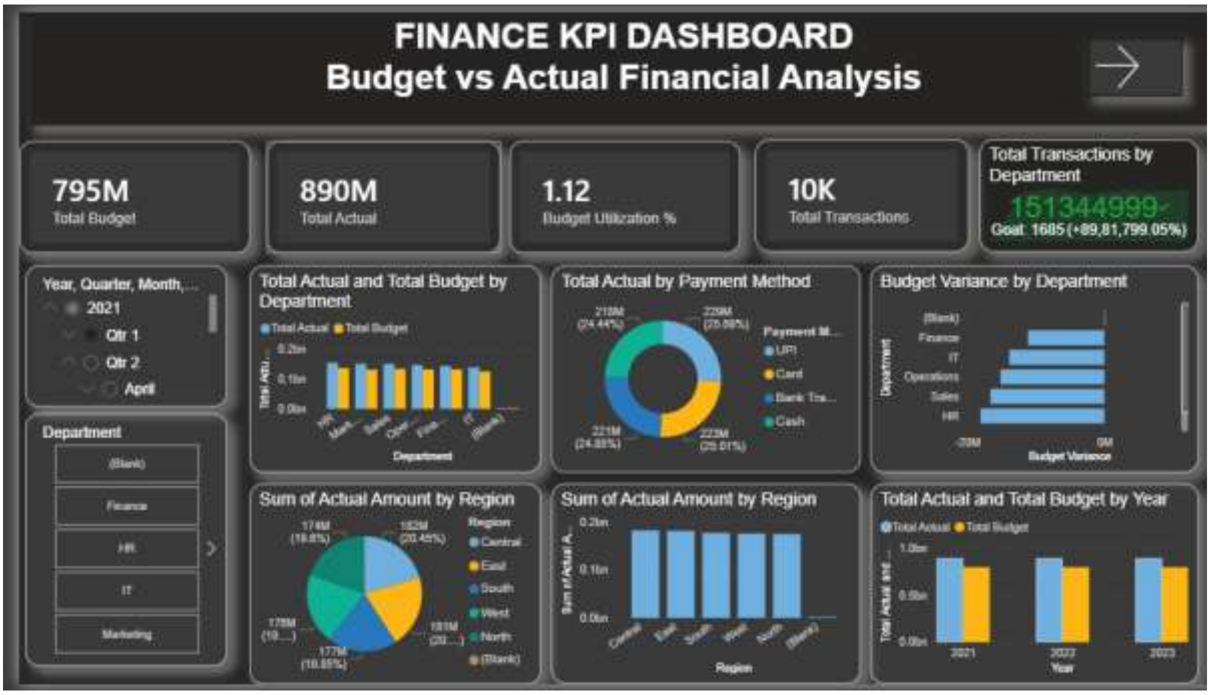
```
Budget Utilization % =
DIVIDE(
    [Total Actual],
    [Total Budget],
    0
)
```

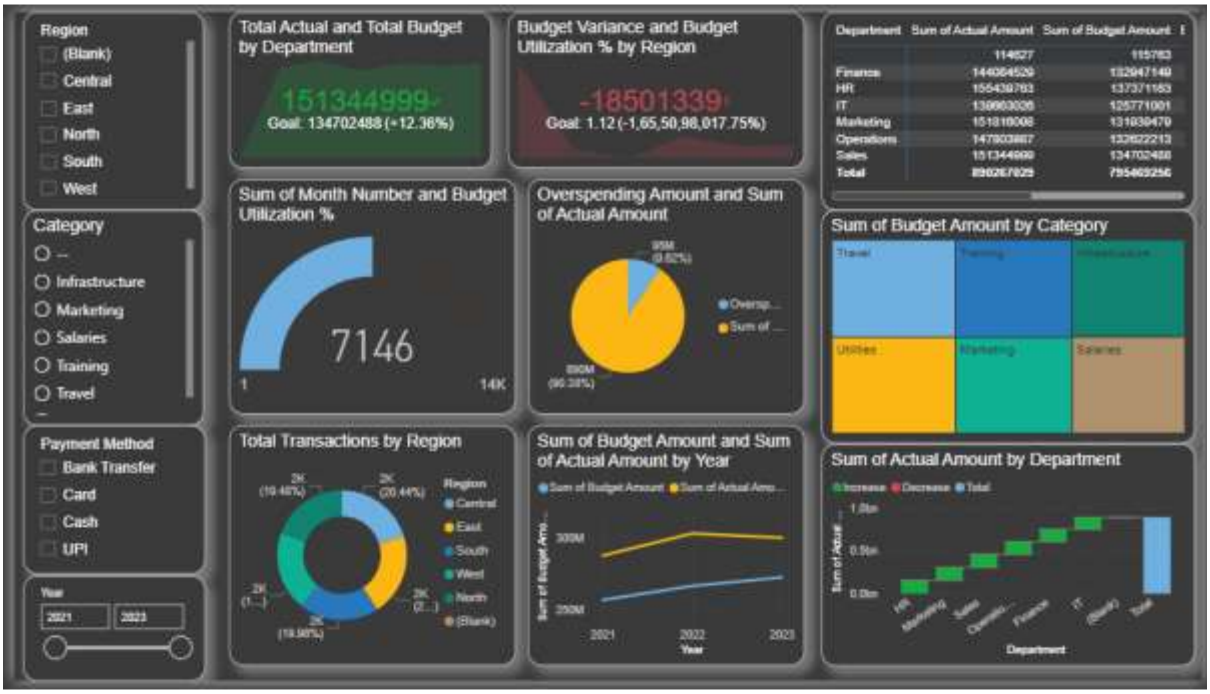
```
Total Transactions =
DISTINCTCOUNT('Financial Data'[Transaction ID])
```

```
Variance % =
DIVIDE(
    [Budget Variance],
    [Total Budget],
    0
)
```

```
Budget Status =
IF(
    [Budget Variance] >= 0,
    "Under Budget",
    "Over Budget"
)
```

```
Overspending Amount =
IF(
    [Budget Variance] < 0,
    ABS([Budget Variance]),
    0
)
```





Results

• The Budget vs Actual Financial Dataset was successfully imported, cleaned, and analyzed using Power BI.

• Data quality was verified by checking data types, duplicate records, missing values, and transaction information.

• KPI Cards provided an overview of total budget, total actual expenditure, budget variance, budget utilization, and total transactions.

• The Department-wise Budget vs Actual chart helped identify differences between planned and actual expenditure across departments.

• The Budget Variance chart helped identify departments with higher or lower financial variance.

• The Category-wise chart helped analyze the distribution of budget amounts across different financial categories.

• The Region-wise charts provided an overview of actual expenditure across different regions.

• The Payment Method chart showed the distribution of actual expenditure across Card, Cash, UPI, and Bank Transfer.

• The Year-wise Budget vs Actual chart helped compare financial performance across different years.

• Interactive slicers allowed users to filter dashboard results by year, department, region, category, and payment method.

• The dashboard showed approximately 795M Total Budget and 890M Total Actual Amount based on the overall dashboard data.

• The Budget Utilization was approximately 112%, indicating that actual expenditure was higher than the allocated budget.

• The dashboard presented financial information in a clear and interactive format, supporting faster and more informed financial decision-making.

Conclusion

This experiment demonstrated how Power BI can be used to design an interactive Finance KPI Dashboard for financial reporting and analysis. The dashboard combines multiple visualizations to monitor budget allocation, compare actual expenditure, analyze budget variance, evaluate department and category performance, and understand regional and payment-method spending patterns.

Interactive filters improve the user experience by allowing customized financial analysis. The dashboard provides valuable insights into budget utilization and spending performance, helping management identify overspending areas and support effective financial planning and decision-making.

Learning Outcomes

✓ Understand the importance of interactive dashboards in financial business intelligence.

✓ Import and prepare the Budget vs Actual Financial Dataset in Power BI.

✓ Verify data quality by checking data types, duplicates, missing values, and transaction records.

✓ Create KPI Cards to summarize financial performance.

✓ Create DAX measures for budget, actual expenditure, variance, utilization, and transactions.

✓ Create Clustered Charts to compare budget and actual amounts.

✓ Create Donut Charts to analyze payment methods and regional expenditure.

✓ Create Bar Charts to analyze department-wise budget variance.

✓ Create Treemaps to analyze category-wise budget allocation.

✓ Create Year-wise charts to monitor financial trends.

✓ Use slicers to build interactive financial reports.

✓ Apply professional color contrast, formatting, and visualization techniques to design an official-looking Power BI dashboard.

✓ Analyze budget utilization and identify potential overspending.

✓ Gain practical experience in developing business intelligence dashboards for financial performance analysis.

Github link: https://github.com/lavanyae4319/Data_Analytics_Assignments